

WASP-135b

R-0033

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_260428 · created 2026-07-08T20:24:13+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461158.823 +/- 0.015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.05
Transit depth (Rp/Rs)^2	3.02 +/- 1.75 [%]
Orbital Inclination (inc)	84.91 +/- 2.93
Airmass coefficient 1 (a1)	0.927 +/- 0.037
Airmass coefficient 2 (a2)	0.073 +/- 0.068
Scatter in the residuals of the lightcurve fit is	3.96 %
Best Comparison Star	None
Optimal Aperture	6.45
Optimal Annulus	10.07
Transit Duration (day)	0.044 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.174 ± 0.05 vs 0.139 (deviation 0.54σ)
Duration fit vs published	0.044 d vs 0.0687 d (deviation 1.13σ)
Mid-time O–C	0.3 min
Depth SNR	2.7
Residual scatter	3.96 %

Light curve

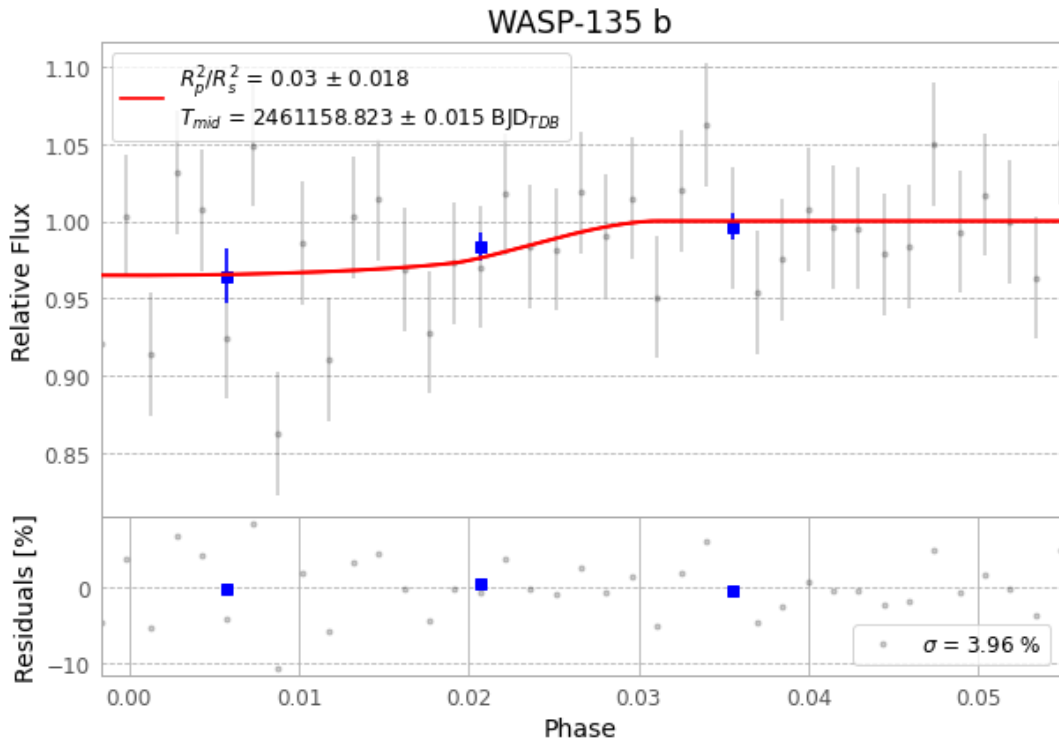


Figure 1 — FinalLightCurve_WASP-135 b_2026-04-28.png (SHA-256 4394e21a2c0a236c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [285, 253], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6944f5be17670480...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

39 FITS frames (26 MB), WASP-135260428073616.FITS → WASP-135260428093016.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-260428.log (b9390d488a96...), FinalLightCurve_WASP-135 b_2026-04-28.png (4394e21a2c0a...), FinalLightCurve_WASP-135 b_2026-04-28.csv (f26841c80e47...)

Compute resources

Wall time 105.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.